The Incidence of "Tampon Taxes": Evidence from the Elimination of State Sales Taxes on Feminine Hygiene Products *

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Abstract

This paper examines the consumer incidence and targeting efficiency of state sales tax exemptions for feminine hygiene products (FHPs). Using NielsenIQ Retail Scanner and Consumer Panel data, we implement a stacked difference-in-differences event-study design to estimate the effects of tax removal on sales quantities and tax-exclusive prices in states adopting these between 2015 and 2020. We find no evidence of price adjustment, implying full pass-through of tax changes to consumers, and a modest increase in quantities sold of approximately 2 percent. Heterogeneity analyses reveal substantially larger demand responses for products primarily purchased by low-income households. However, back-of-the-envelope calculations indicate that an expenditure-equivalent, means-tested cash transfer would provide 4 to 10 times more benefit per recipient than a broad-based tax exemption. These findings highlight important trade-offs between equity, efficiency, and targeting in the design of consumption tax policy.

Keywords: Tax Incidence, Sales Tax, Feminine Hygiene Products, Stacked Difference-in-Difference

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1 Introduction

Nearly all U.S. states with a retail sales tax have either eliminated or considered eliminating its application to feminine hygiene products (FHPs). Supporters of the so-called "tampon tax" exemptions offer multiple rationales that combine traditional concerns about sales tax regressivity with arguments related to gender equity and basic human needs. As with other goods commonly classified as necessities, such as groceries or pharmaceuticals, proponents argue that FHPs should not be taxed in the same manner as non-essential or luxury items. This perspective reflects the standard view that consumption taxes can impose disproportionate burdens on low-income households, alongside a more recent argument that excluding FHPs from such exemptions overlooks issues of menstrual or "period poverty" (Crawford and Spivack, 2017). Survey evidence is often cited in support of these claims. For example, a survey of 1,000 U.S. teens aged 13 to 19 who menstruate reports that one in five has struggled to afford or access period products (Thinx and PERIOD, 2019).[1] Advocates further warn that "an added tax burden on necessary hygiene products forces many to use menstrual products longer than the recommended six to eight hours," increasing the risk of serious health conditions such as Toxic Shock Syndrome and limiting students' ability to perform at their full potential in school and workplace performance (Arumilli et al., 2020).

The purpose of this research is to inform state policy decisions regarding the targeting efficiency of sales tax exemptions for feminine hygiene products (FHPs). Proposed exemptions typically require fiscal notes estimating their revenue impact during their legislative consideration, and many states mandate periodic review of tax expenditures for evidence of effectiveness. To contribute to this evaluation, we examine three related questions: (1) Does eliminating state sales taxes on FHPs increase their unit sales? (2) To what extent is the tax exemption passed through to consumer prices? (3) How does this policy compare with an expenditure-equivalent means-tested cash transfer to low-income females of menstruating age? The first two questions identify behavioral responses and tax incidence, while the third evaluates the targeting efficiency of the policy relative to a direct transfer.

We use NielsenIQ Retail Scanner and Consumer Panel Survey data from January 2015 to

[1]The survey was sponsored by Thinx, a New York based manufacturer of feminine hygiene products, in collaboration with PERIOD, a non-governmental organization.

January 2020. Over this period, six states adopt an FHP exemption with at least 12 months of pre- and post-treatment data surrounding their adoption month. We estimate the effects of these policies on the sale quantities and tax-exclusive prices at the UPC-by-retail-store level in a stacked differences-in-differences (DiD) framework with event-time indicators, comparing treated states to a set of never-adopting controls. Using the Consumer Panel Survey of households, we classify a subset of FHP products according to whether they are primarily purchased by low- or high-income households. We then re-estimate our stacked DiD specification separately on these product groups. The high frequency nature of the data yields precise estimates of both demand and pricing responses. We also examine heterogeneity by retailer and geographic characteristics.

To preview our findings, we find that sales tax exemptions increase the quantity of FHP units sold while having no effect on retailers' tax-exclusive prices. This pattern is consistent with full pass-through of the tax savings to consumers. The estimated increase in quantity is approximately 2.1 percent; combined with an average waived sales tax rate of 8.1 percent, this implies a tax-price elasticity of demand of -0.247. We also see elasticities that are about five times larger for products predominantly sold to low-income households. Taken together, these results suggest that the broad FHP sales tax exemptions transfer resources to consumers and modestly expand access to these products.

Sales tax exemptions for FHPs transfer state resources to all purchasers of these products, rather than only to the populations policymakers intend to target. Using American Community Survey data, we estimate the annual fiscal cost of these exemptions and calculate the per-recipient benefit under an expenditure-equivalent, means-tested cash transfer program targeted by income and age. Our estimates suggest that such a targeted program could increase benefits per recipient by a factor of 4 to 10 at the same fiscal cost to the state.

This paper contributes to the nascent tax incidence literature on "tampon taxes." A small number of studies examine the effects of removing sales taxes on menstrual hygiene products (FHPs) in specific U.S. contexts. In the American setting, Cotropia and Rozema (2018) use NielsenIQ Consumer Panel data and a difference-in-differences design to study the repeal of the tampon tax in New Jersey in 2005. Dividing households by income, the authors find that low-income consumers receive benefits exceeding the size of the repealed tax, while high-income consumers

share the tax reduction more evenly with producers. Yang (2025a) also uses NielsenIQ Consumer Panel data but examines Florida's 2018 repeal using synthetic controls. The study finds that unit prices of menstrual hygiene products declined following the repeal, with larger price reductions for low-income than for high-income consumers. The case of Texas' 2023 repeal is examined in Yang (2025b) for impacts on consumer and producer surplus, with much of the gains coming from lower income households. Outside the United States, both Frey and Haucap (2024) and Buettner et al. (2025) examines a value-added tax (VAT) reduction for menstrual hygiene products in Germany and finds that the tax reduction was fully passed through to consumers with limited demand responses.

This paper makes three contributions to the literature on the incidence and policy design of sales taxes on feminine hygiene products (FHPs). First, we provide the first multi-state evidence on the effects of eliminating these taxes, leveraging variation across six U.S. states. Existing studies focus on single-state reforms, limiting external validity and the ability to assess heterogeneity across policy environments. Second, we use high-frequency NielsenIQ Retail Scanner data in a stacked difference-in-differences and event-study framework at the UPC-by-retail-store level, allowing for rich fixed effects and precise estimation of both demand responses and tax incidence. This approach enables us to separately identify changes in quantities and tax-exclusive prices, providing direct evidence on pass-through. Third, we evaluate the targeting efficiency of FHP tax exemptions by comparing their implied benefits to those of an expenditure-equivalent, means-tested cash transfer. This comparison highlights the extent to which broad-based tax exemptions diffuse benefits beyond the intended population and quantifies the potential gains from more targeted policy instruments.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and data, Section 3 outlines the empirical strategy, Section 4 presents the results, Section 5 evaluates targeting efficiency, and Section 6 concludes.

2 Institutions and Data

2.1 Staggered Policy Adoption in the States

State sales tax systems in the United States rely heavily on product-specific exemptions.[2] The retail sales tax, which originated in the 1930s, was designed to tax "finished" goods and services rather than the broader concept of final consumption typically emphasized by tax economists. As a result, many transactions within the business-to-business production chain are subject to taxation, creating the potential for tax pyramiding, unless explicitly exempted. In addition, policymakers have introduced exemptions motivated by concerns about regressivity and the taxation of necessities, with groceries and pharmaceuticals among the most prominent examples. Together, these features generate a tax base defined as much by exemption as by inclusion, producing numerous boundary cases and raising concerns about horizontal equity. For example, the tax treatment of similar goods may vary depending on classification: items such as soda or prepared foods may or may not qualify as exempt groceries depending on how they are sold, and identical products may be taxed differently when purchased by businesses versus households. More broadly, exemptions are also used as policy instruments to encourage specific economic activities or sectors.

The tax treatment of FHPs reflects the broader challenges of defining the sales tax base that is readily identified in the public discourse. Under standard statutory language, these products are taxable as "finished goods" unless covered by another exemption. For example, Minnesota exempted FHPs in 1981 as part of a broader exemption for health-related products, while Pennsylvania excluded them in 1991 through a sales tax repeal on paper products. By contrast, Indiana exempted certain medical devices that collect bodily fluids (e.g., colostomy or ileostomy bags) in 2011, but this exemption was not interpreted to include bandages or FHPs. For most states, FHPs remained part of the general sales tax base, and prior to 2000 no state enacted an exemption specifically targeting menstrual hygiene products. Beginning in the 2010s, however, states increasingly adopted legislation explicitly exempting these products from sales tax. Figure 1 summarizes the policy landscape as of December 2025. Twenty U.S. states continue to impose sales taxes on FHPs,

[2]The reference material for this paragraph can be found in Mikesell et al. (2023) and Mikesell and Ross (2025).

while others either do not levy a general sales tax or have removed these taxes through legislative reforms. This variation across states and over time provides a natural setting to evaluate the incidence and targeting efficiency of these exemptions.

2.2 Data

Retail scanner data. The primary data source for this analysis is the NielsenIQ Retail Scanner Dataset (Kilts Center Archive of the Neilson Company, 2019). These data provide weekly information on prices, sales volumes, and in-store merchandising conditions collected from point-of-sale systems of participating retail stores across U.S. markets, excluding Hawaii. The dataset covers approximately 30,000 to 35,000 grocery stores, drug stores, mass merchandisers, and other retail outlets. While participation is not universal, the dataset captures a large share of retail sales in covered sectors and is widely used in empirical work on consumer demand and pricing. The data include products across all categories tracked by NielsenIQ, including food, non-food grocery items, health and beauty products, and selected general merchandise. In addition to weekly product movement files, the dataset contains annual store-level files and detailed product information files. Merging these components allows us to construct measures of sales quantities and prices at the UPC-by-store-by-week level, which form the basis of our empirical analysis.

This paper uses NielsenIQ grocery movement files for the following product categories from January 2015 to January 2020: tampons, sanitary napkins, and sanitary belts and panties.[3] For each UPC, we construct store-by-week measures of total units sold and the tax-exclusive mean price, which serve as the primary outcome variables in our analysis.

Consumer panel data. We also use NielsenIQ Consumer Panel data to classify products according to the income groups that primarily purchase them, which we use to examine heterogeneity in treatment effects. The NielsenIQ Consumer Panel is a longitudinal dataset that tracks the purchasing behavior of a representative sample of U.S. households, providing detailed information on household demographics and product-level purchases.

[3]Although menstrual cups are included in some post-repeal exemptions, we exclude them because they are classified as "unassigned feminine hygiene products" in the dataset. Since UPC-level product descriptions are limited, we risk including products not-exempted by the policy by including all unassigned FHPs. However, menstrual cups represent a relatively small share of the market compared to disposable menstrual products.

Sales tax rate history. We compile state sales tax rate histories from the Federation of Tax Administrators (FTA), which provides detailed information on state-level sales tax rates, bases, and statutory changes over time. The FTA data include the timing of legislative changes to general sales tax rates. For the FHP exemptions, we validate state policies identified by the Alliance for Period Supplies as part of their tampon tax project. These measures are merged with our retail scanner data based on state and time, forming the basis for our treatment definitions in the difference-in-differences and event-study analysis.

State demographics on female poverty and age. To estimate the fiscal cost of the tax expenditure by state, as well as the cost of an equivalent means-tested cash grant, we rely on data on female populations by age and poverty status from the 2023 American Community Survey 1-Year Estimates, specifically Table S1701 on poverty status by state and Subject Table S0101 on age and sex.

3 Empirical Model

Our primary econometric approach is a stacked difference-in-differences design, which addresses bias arising from the staggered adoption of the FHP tax exemption across states over time. Section 3.1 defines the treatment and control groups, section 3.2 presents the stacked DiD specification, and section 3.3 examines product-level heterogeneity.

3.1 State Selection

Our treatment group consists of states that adopted sales tax exemptions for FHPs between 2016 and 2019, allowing for a balanced event-study window of two years with sufficient pre-trends and post-treatment dynamics. We exclude states adopting after January 2019 to avoiding confounding effects from the COVID-19 pandemic that began in March 2020, and earlier adopters due to limited pre-period coverage in the scanner data. This yields six states across five adoption cohorts: New York (September 2016), Illinois (January 2017), Connecticut (January 2018), Florida (January 2018), Maryland (July 2018), and Nevada (January 2019). The timing of these policy changes is summarized in the map in panel b of Figure 1.

The control group consists of states that do not adopt FHP sales tax exemptions during our event window. This definition excludes early adopters (Pennsylvania, Massachusetts, Minnesota, and New Jersey) as well as Rhode Island, which adopted in October 2019, from the control group. We also exclude states with incomplete scanner data coverage, including Alaska, Hawaii, and Tennessee. For each state in the analysis, we retain a balanced window of 12 months before and after the cohort adoption month.

3.2 Stacked DiD

Our unit of observation is a product-store pair, where FHP products are defined by UPC codes and stores correspond to physical retail locations. The combination of UPC code i and store j uniquely identifies an observational unit, which we observe weekly and collapse to the monthly level.

To address concerns arising from staggered adoption of FHP tax exemptions, we employ a stacked dynamic event study and difference-in-difference framework. For product i, store j, state s, sub-experiment cohort c, and month t we estimate:

$$ln(Y_{ijsct}) = \alpha_{ij} + \sum_{\tau=-12}^{-2} \rho_\tau 1[t - T_c = \tau] \cdot Treat_c + \sum_{\tau=0}^{12} \nu_\tau 1[t - T_c = \tau] \cdot Treat_c + \gamma_{sc} + \delta_{tc} + \epsilon_{ijsct} \quad (1)$$

The outcome Y is either the total units sold or the mean price of the product. Units are defined as total items sold, adjusting for package size (e.g., a 12-count product multiplied by units sold). Prices are constructed as sales-weighted average pretax prices within each UPC-store-month cell. $Treat_c$ equals 1 for treated observations in cohort c.

Event time is defined as $\tau = t - T_c$, where T_c denotes the adoption month for the cohort. The specification includes indicators for $\tau \in \{-12, \ldots, -2, 0, \ldots, 12\}$, omitting $\tau = -1$ as the reference period. The fixed effects $\alpha_{i,j}$, $\gamma_{s,c}$, and $\delta_{c,t}$ denote UPC-by-store, state-by-cohort, and cohort-by-month effects, respectively. Under this specification, the coefficients on the event-time indicators capture the dynamic effects of the policy relative to the period immediately preceding adoption ($\tau = -1$). Event time coefficients for $\tau < 0$ provide a test of the parallel trends assumption ($\rho_\tau = 0$), while post-period coefficients for $\tau \geq 0$ measure the evolution of treatment effects following policy implementation. Identification comes from within-product-store variation over time realtive to

cohort-specific adoption timing.

When price is the outcome, estimates of $\nu_\tau = 0$ imply that retailers do not adjust their tax-exclusive price following the adoption of the FHP tax exemption, consistent with full pass through of the tax savings to consumers. Positive estimates ($\nu_\tau > 0$) would indicate some recapture of the tax exemption by the retailer through higher pretax prices. When quantity is the outcome, positive post-treatment coefficients ($\nu_{tau} > 0$) indicate increased consumption in response to the reduction in tax-inclusive prices.

In addition to the dynamic event-study specification in equation (1), we estimate the following stacked difference-in-difference model to obtain a summary treatment effect:

$$ln(Y_{ijsct}) = \beta_0 + \beta_1 \cdot Treat_c \cdot Post_{ct} + \alpha_{ij} + \gamma_{sc} + \delta_{ct} + \epsilon_{ijsct} \tag{2}$$

Standard errors in both model specifications are clustered at the state-by-cohort level. The final full sample consists of 365,279,745 product-by-store-by-month-by-cohort observations on sales quantities and prices. Descriptive statistics are reported in Appendix Tables B.1 and B.2.

3.3 Product Heterogeneity

We estimate our event-study and difference-in-differences specifications on selected subsamples of the full dataset with the goal of better understanding the distributional implications of the policy.

In particular, we examine heterogeneity across products that disproportionately serve lower- and higher-income consumers. The main analysis includes over 2,000 unique UPCs observed across more than 42,000 retail stores. Although the policy applies uniformly at the product level rather than being means-tested by consumer income, product differentiation and market segmentation imply that different brands may serve different segments of the income distribution. Cha and Park (2019) document that higher-end menstrual products tend to emphasize sustainability (e.g., biodegradable or organic materials), health attributes (e.g., hypoallergenic, chemical-free), and social responsibility, whereas lower-priced products emphasize basic functionality. This segmentation approximates income targeting and provides a mechanism through which tax changes may generate heterogeneous effects across income groups.

To operationalize this distinction, we use the 2017 NielsenIQ Consumer Panel to classify products based on the income distribution of purchasing households. We divide households into those above and below the federal poverty line.[4] We then identify products that are exclusively observed to be purchased by households below the poverty line and those exclusively purchased by households above it. This yields 18 UPCs in the below-poverty group and 168 UPCs in the above-poverty group. Consistent with this classification, average monthly expenditure per household is $25.86 for these UPC products purchased by below-poverty households and $31.01 for products purchased by above-poverty households.[5]

4 Regression Results

The results of estimating the dynamic stacked event study in Equation (1) are displayed with 95 percent confidence intervals in Figure 2, with the summary DiD estimates provided in columns (1) and (4) of Table 1. Despite clustering standard errors at the state-by-cohort level, the estimates are sufficiently precise to rule out substantively large effects.

Column (1) of Table 1 and panel (a) of Figure 2 indicate that sales of FHPs increase by approximately 2.1 percent following exemption from the sales tax base, and this effect is statistically significant. The dynamic event-study estimates show a modest initial increase relative to the pre-period, followed by a gradual rise over the subsequent months, with the largest effects occurring toward the end of the first post-treatment year. In contrast, prices exhibit no economically or statistically significant response. As shown in panel (b) of Figure 2, the largest point estimate within a 95 percent confidence interval in any given month is approximately 0.02, and the pooled stacked difference-in-differences estimate implies an increase of 0.1 percent, which is not statistically distinguishable from zero. These results indicate that tax-exclusive prices do not adjust in response to the exemption, consistent with full pass-through of the tax change to consumers. Accordingly, the change in tax-inclusive prices is driven almost entirely by the removal of the sales tax itself,

[4]NielsenIQ categorizes households into income bins; we classify households in bin 13 or lower as low-income. Bin 13 corresponds to annual income below $25,000, approximately the federal poverty threshold for a household of four during the study period.

[5]Although total monthly expenditures are lower for below-poverty households for their respective products, their per item prices are higher on average ($0.37 versus $0.27).

with an average rate of 8.1 percent. Combining this price change with the estimated increase in quantities implies a tax-price elasticity of demand of approximately -0.259.[6]

We next examine heterogeneity across products predominantly purchased by households above and below the poverty line, reported in columns (2), (3), (5), and (6) of Table 1 and in the event-study estimates shown in Figure 3. For products primarily purchased by below-poverty households, we estimate a 10.8 percent increase in units sold following the tax exemption, statistically significant at the 5 percent level. This implies a tax-price elasticity of demand of approximately $(0.108/-0.081) = -1.33$, substantially larger in magnitude than estimates for the full sample. This pattern is consistent with greater price sensitivity among lower-income consumers, as menstrual products likely represent a larger share of their household budgets, leading to higher demand elasticities. In contrast, for products primarily purchased by above-poverty households, we estimate a statistically significant decline in sales of approximately 2.2 percent. The event-study estimates suggest that this effect is driven by short-run fluctuations around event time $t = 3$ that attenuates in later periods, whereas the increase for below-poverty products appears more persistent over time that could be consistent with discovery of the lower tax. We find no evidence of economically meaningful price responses in either group. Although the estimates for below-poverty products are noisier in the pre-treatment period, point estimates remain small and statistically indistinguishable from zero throughout.

The Appendix reports additional heterogeneity analyses in Table B.3, which do not materially alter the main findings. These include variation by store type (e.g., convenience stores, mass merchandisers, drug stores, and food stores) and by store location (urban versus rural counties). Corresponding event-study plots and regression results are presented in Figures B.2 and B.3.

5 Estimates for a Targeted Assistance Alternative

There are multiple motivations for exempting feminine hygiene products (FHPs) from sales taxation. A central rationale is that these products are necessities, and taxing them imposes a disproportionate burden on lower-income households. In this sense, FHP exemptions are often framed as

[6]The tax elasticity calculation is $\frac{0.021}{-0.081} = -0.259$.

a corrective to the regressivity of consumption taxation, particularly for goods with limited substitutability and recurring demand. However, broad-based tax exemptions are not the only policy response to concerns about affordability and access. Policymakers frequently consider alternative, expenditure-based approaches that provide more targeted support. For example, in the United Kingdom, revenues from the value-added tax on sanitary products are allocated to a fund supporting nonprofit organizations that assist low-income women by purchasing and distributing menstrual products to individuals in need, illustrating a policy design that concentrates benefits on a specific population rather than distributing them across all consumers.

These two approaches—broad-based tax exemptions and targeted transfers—differ fundamentally in their ability to direct resources toward intended beneficiaries. While tax exemptions reduce prices for all consumers, they may deliver substantial benefits to higher-income households that would have purchased these products regardless of price. In contrast, targeted programs can concentrate resources on individuals for whom affordability constraints are most binding, albeit at the cost of administrative complexity. The empirical analysis that follows evaluates both the behavioral responses to tax exemptions and their implied distributional consequences, and compares these outcomes to those from an expenditure-equivalent, means-tested transfer.

To quantify this trade-off, we compare the revenue loss from a sales tax exemption to a budget-equivalent, means-tested cash transfer. A tax exemption generates savings for all consumers regardless of income, whereas a targeted transfer allocates the same fiscal resources exclusively to lower-income households.[7] Aggregating forgone tax revenue across all purchases, we construct a measure of tax expenditure for each state s:

$$TE_s = F_s \Delta\tau_s PQ.$$

Where F_s denotes the number of menstruating individuals, $\Delta\tau_s$ is the waived general sales tax rate, P is the average price per unit, and Q is the average annual quantity consumed per individual.

The average per-recipient benefit under the tax exemption is given by

[7]See Appendix section A for more detail and rationale of assumptions used in calculations.

$$TE_s^F = \frac{TE_s}{F_s},$$

while the corresponding benefit under a revenue-equivalent, means-tested transfer targeted to individuals below the poverty line (F_s^P) is

$$G_s^{FP} = \frac{TE_s}{F_s^P}.$$

Comparing TE_s^F and G_s^{FP} highlights the targeting inefficiency of broad-based tax exemptions. Variation across states arises from differences in sales tax rates and the share of the population below the poverty line. Because we assume constant prices and quantities across individuals, these calculations may modestly overstate benefits for lower-income households to the extent that demand exhibits positive income elasticity.

Appendix Table B.4 presents these calculations for all 45 states with a sales tax. Figure 4 reports results for the six treated states from the DiD study, with green bars indicating per-recipient savings under the tax exemption and purple bars showing the equivalent targeted transfer. On average, the tax exemption provides approximately $4.39 in annual savings per menstruating individual, whereas the targeted transfer provides more than eight times as much at $35.62. Extending the analysis nationally, the ratio ranges from 4.74 in Louisiana to 10.53 in Utah.

Overall, these results indicate that a revenue-equivalent, poverty-targeted transfer would increase per-recipient benefits by a factor of approximately 4 to 10 relative to a broad-based sales tax exemption. One interpretation of these findings is that a UK-style policy earmarking sales tax revenues from FHPs could finance approximately one month of product supply for individuals living in low-income households.

6 Conclusion

This paper studies the consumer tax incidence of state sales tax exemptions for feminine hygiene products using NielsenIQ Retail Scanner and Consumer Panel data. We find no effect of the tax exemption on tax-exclusive prices and a modest increase in quantities sold. Examining a subset of

products predominantly purchased by households below the poverty line, we estimate an increase in sales of approximately 10 percent, with no evidence of retailer price adjustment. These results imply that the tax benefit is fully passed through to consumers and is associated with an expansion in access. However, comparing the implied tax savings to a revenue-equivalent, means-tested cash transfer, we find that a targeted program would deliver substantially larger benefits. Specifically, a transfer to low-income individuals of menstruating age would provide approximately 4 to 10 times more benefit per recipient at the same fiscal cost to the state. These findings inform policy discussions regarding the welfare and distributional trade-offs between broad-based tax exemptions and more targeted program alternatives.

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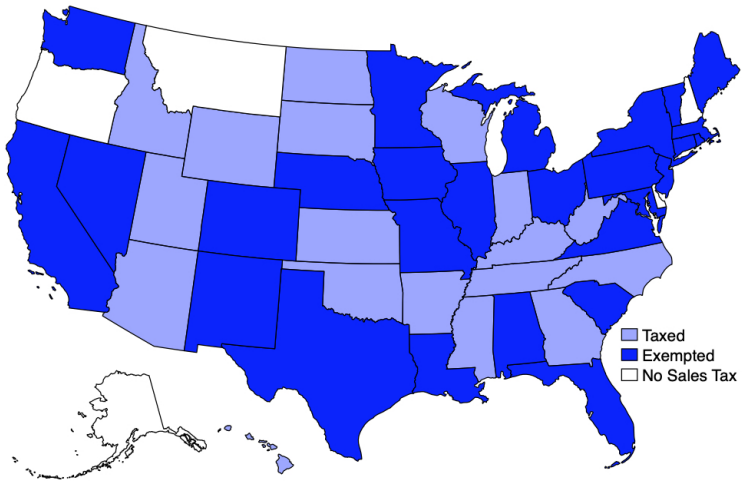
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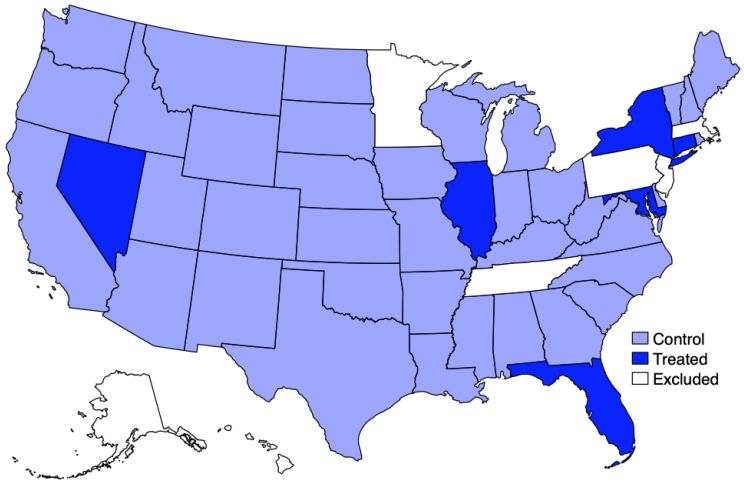
Figures & Tables

Figure 1: Sales Tax Status on Feminine Hygiene Products in the United States

(a) Sales Tax on Feminine Hygiene Products by State (Dec 2025)



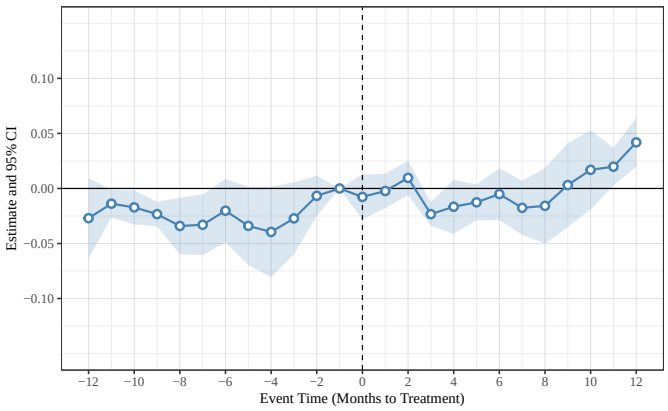
(b) States in Treated and Control Groups



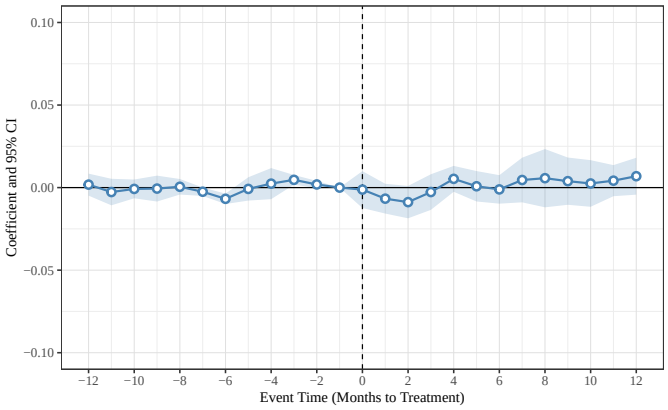
Notes: This figure illustrates the policy background and sales tax status of feminine hygiene products collected from various sources that include Federation of Tax Administrators and the Alliance for Period Supplies. Treated states in panel (b) are the subset of exempt states in panel (a) that adopted the FHP exemption during the 2016 to 2019 event window.

Figure 2: Effects of Elimination of State Sales Taxes on Feminine Hygiene Products

(a) Effects on ln(Quantity)



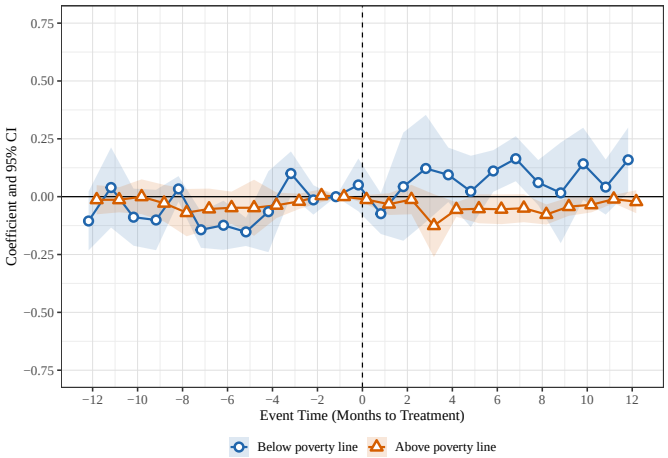
(b) Effects on ln(Price)



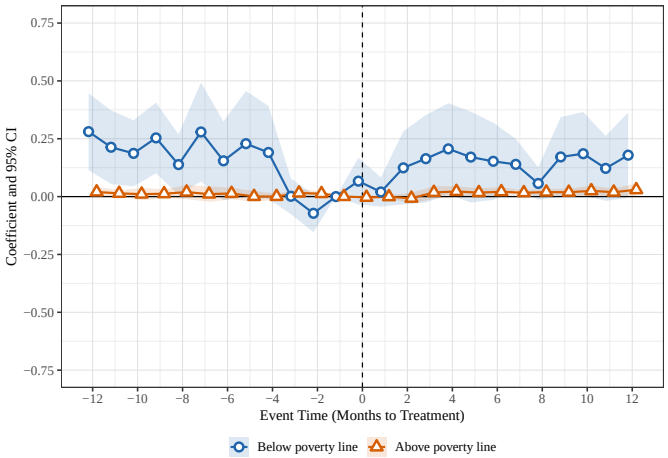
Notes: This figure shows the event study estimates of the policy's impact on feminine hygiene products sales. Panel (a) displays the effects on sales quantity and panel (b) displays the effects on sales prices.

Figure 3: Effect of Elimination of State Sales Taxes on Feminine Hygiene Products by Product Groups

(a) Effects on Quantity

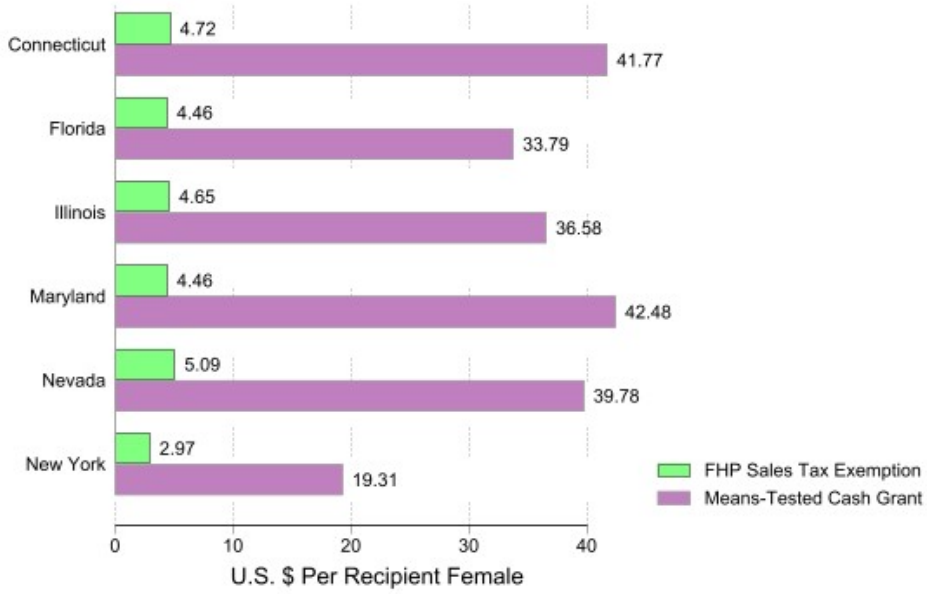


(b) Effects on Price



Notes: This figure shows the event study estimates of the policy's impact on feminine hygiene products sales by product groups. Panel (a) displays the effects on sales quantities and panel (b) displays the effects on sales prices by product groups. In both Panel (a) and (b), the estimates generated from the sample of products purchased households below the poverty line are in blue with circle markers, and those above the poverty line are in vermilion with triangle markers.

Figure 4: Per Recipient Comparison of FHP Exemption Equivalent Poverty Targeted Cash Grant



Notes: This figure shows the estimated tax savings per menstruating female under the sales tax exemption (green) and a targeted cash transfer of equivalent revenue cost to menstruating females below the poverty line (purple). Methodology is described in section A, and estimates for all 45 sales taxing states found in Table B.4.

Table 1: Stacked DiD Results for State FHP Sales Tax Exemption

	ln(Quantity)			ln(Price)		
	All	Below FPL	Above FPL	All	Below FPL	Above FPL
	(1)	(2)	(3)	(4)	(5)	(6)
Treat × Post	0.021**	0.108**	-0.022**	0.001	0.003	0.002
	(0.009)	(0.043)	(0.010)	(0.005)	(0.015)	(0.005)
State×Cohort FE	Y	Y	Y	Y	Y	Y
Time×Cohort FE	Y	Y	Y	Y	Y	Y
UPC×Store FE	Y	Y	Y	Y	Y	Y
Observations	365,279,745	68,274	5,834,395	365,279,745	68,274	8,778,596

Notes: Robust standard errors clustered by state and cohort reported in parentheses. Statistical significance indicated at the 0.01 (∗ ∗ ∗), 0.05 (∗∗), and 0.10 (∗) level. Outcomes are the logged quantity of sales (columns 1-3) and price (columns 4-6) as described in section 3. All FHP products are included in Columns (1) and (4); columns (2) and (5) are products only purchased by households below the poverty line in the 2017 Consumer Panel; Columns (3) and (5) are products only purchased by households above the poverty line. Treatment and control states listed on Figure 1.

A Sources and Methodology for Targeted Assistance Alternative

To parameterize menstrual product usage, we draw on estimates from Menstruation Facts and Figures produced by the Association of Hygiene Product Manufacturers (AHPMA).[8] These estimates imply approximately 22 products per menstrual cycle and 13 cycles per year, with menstruation beginning at age 12.5 and continuing through age 50. Using a mean per-unit price of \$0.26 from NielsenIQ Retail Scanner data for 2023, these inputs imply an annual pre-tax expenditure of \$74.33 per menstruating individual. The regression results in section 4 found no evidence of price adjustments by retailers in response to the exemptions, so we keep the price unchanged across the two programs.

State general retail sales tax rates for 2023 are obtained from the Federation of Tax Administrators and are used to approximate the waived tax rate under FHP exemptions ($\Delta\tau_s$). If the general sales tax rate in the state was 6 percent, then waiving it on pre-tax expenditures of \$74.33 would represent about \$4.46 in annual tax savings.

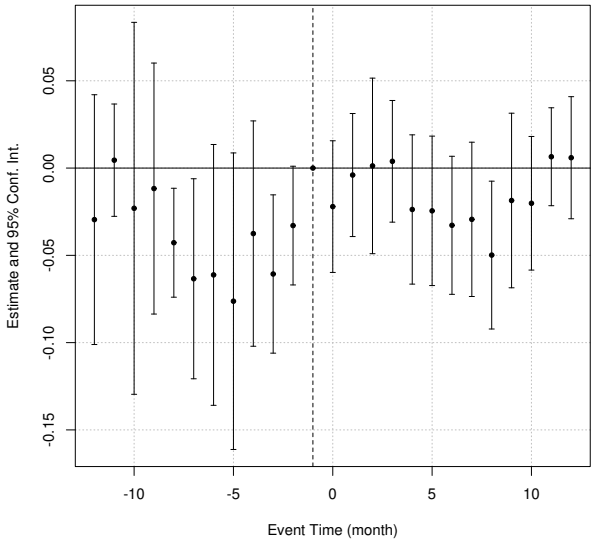
State population estimates are drawn from the 2023 American Community Survey (ACS) 1-Year Estimates. Because ACS data are reported in age bins, we approximate the number of menstruating individuals (F_s) as the sum of females aged 15–49, plus 50 percent of those aged 10–14 and 20 percent of those aged 50–54. We estimate the number of menstruating individuals below the poverty line (F_s^P) by applying state-level female poverty rates from the ACS to these population counts.

[8] `https://www.ahpma.co.uk/menstruation_facts_and_figures/`

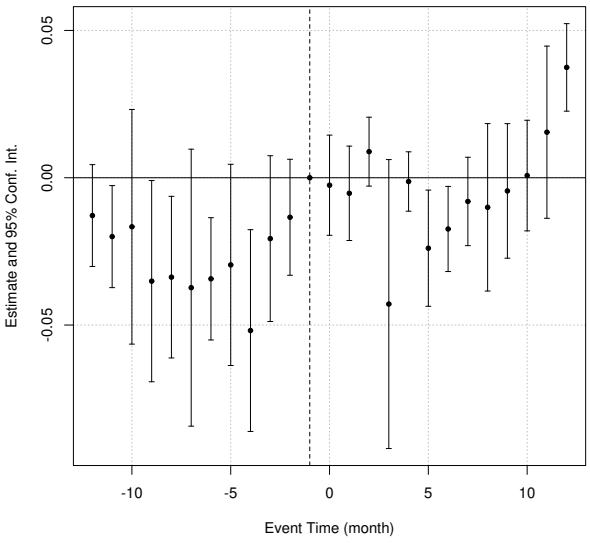
B Figures and Tables

Figure B.1: Effects of Elimination of State Sales Taxes on Feminine Hygiene Products Sales Quantity by Store Types

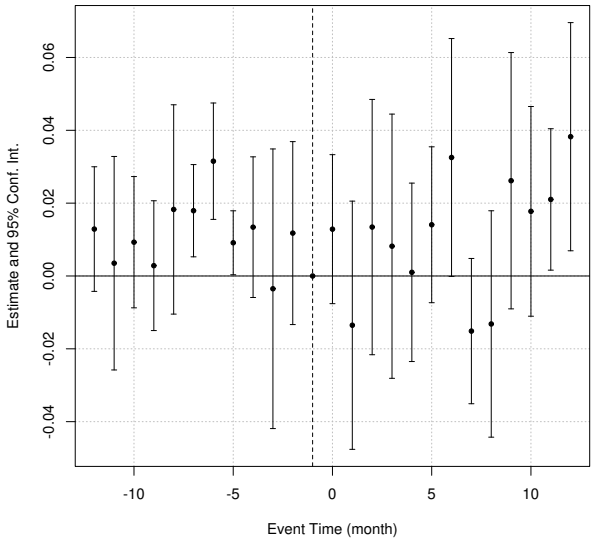
(a) Convenience Stores



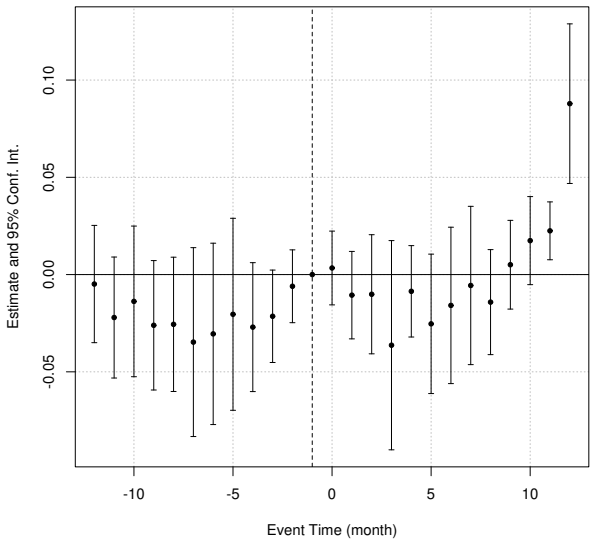
(b) Drug Stores



(c) Food Stores



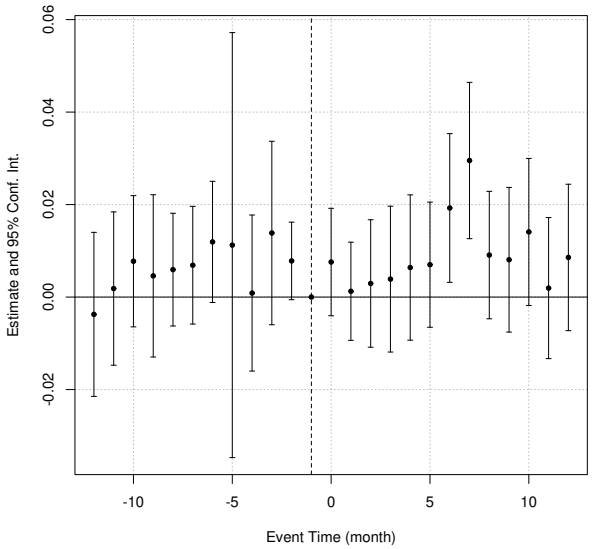
(d) Mass Merchandisers



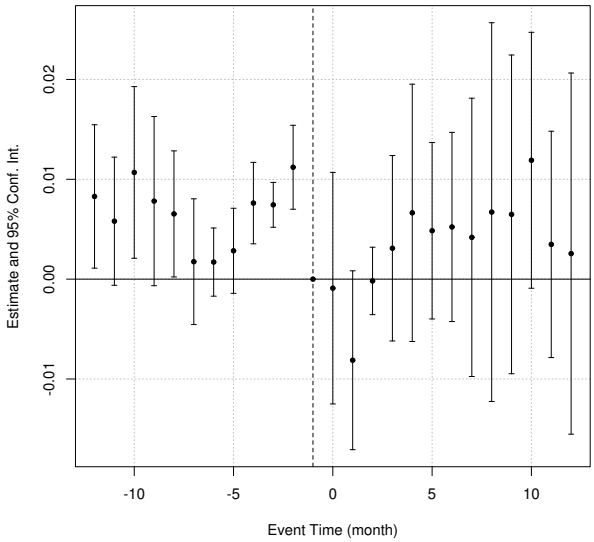
Notes: This figure presents the event study estimates of the policy's impact on the sales quantity of feminine hygiene products, disaggregated by store type. Panels (a), (b), (c), and (d) report the results for convenience stores, drug stores, food stores, and mass merchandisers, respectively.

Figure B.2: Effects of Elimination of State Sales Taxes on Feminine Hygiene Products Sales Price by Store Types

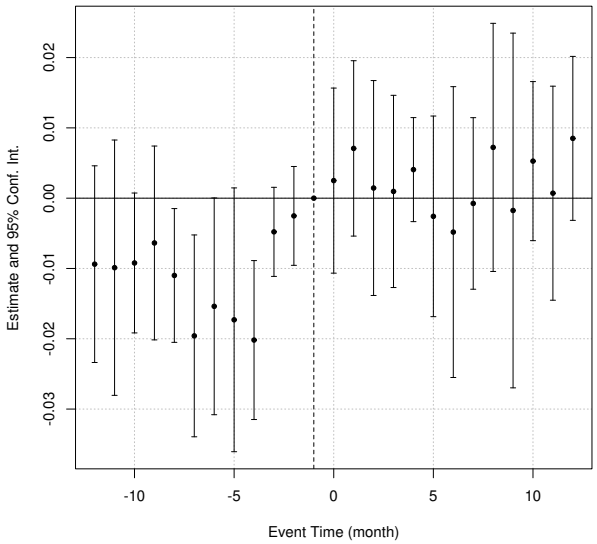
(a) Convenience Stores



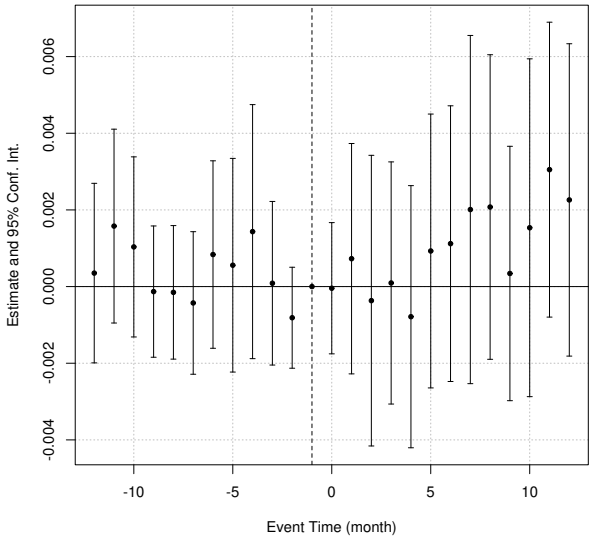
(b) Drug Stores



(c) Food Stores

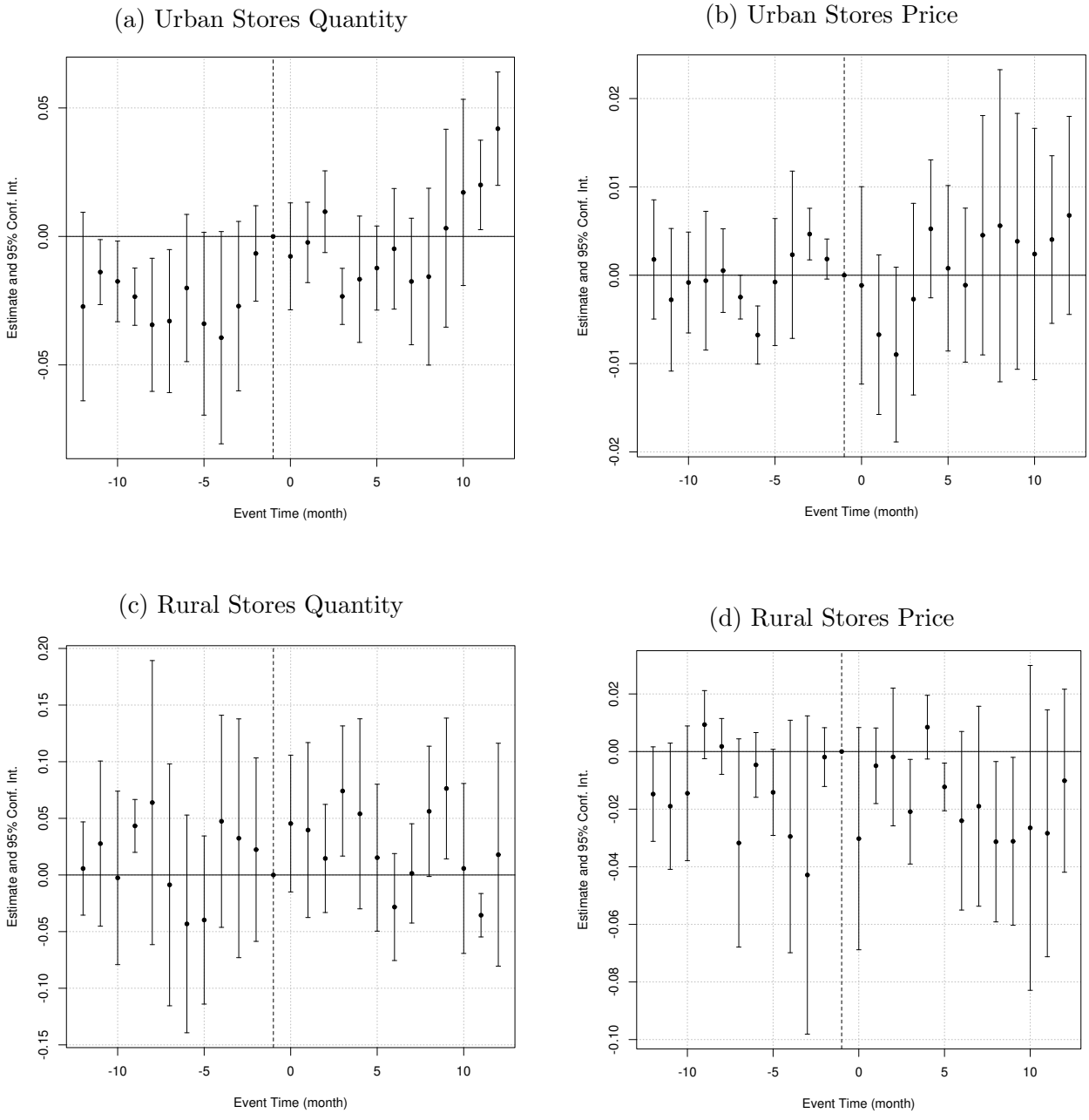


(d) Mass Merchandisers



Notes: This figure presents the event study estimates of the policy's impact on the sales prices of feminine hygiene products, disaggregated by store type. Panels (a), (b), (c), and (d) report the results for convenience stores, drug stores, food stores, and mass merchandisers, respectively.

Figure B.3: Effects of Elimination of State Sales Taxes on Feminine Hygiene Products Sales Quantity and Price by Store Locations



Notes: This figure presents the event study estimates of the policy's impact on the sales quantity and prices of feminine hygiene products, disaggregated by store location. Panels (a), (b), (c), and (d) report the results for urban store sales quantity, urban store sales price, rural store sales quantity, and rural store sales price, respectively.

Table B.1: Summary Statistics of Feminine Hygiene Products

	Treated Group	Control Group	Total
UPCs			
All Products	1740	2071	2141
Low-income only Products	15	17	18
High-income only Products	155	166	168
States	6	35	41
Counties			
Urban Counties	247	1797	2044
Rural Counties	5	308	313
Store Numbers			
All Channels	8963	33842	42805
Convenience Stores	1933	8277	10210
Drug Stores	3046	7432	10478
Food Stores	1823	9104	10927
Mass Merchandiser	2161	9029	11190
Urban Stores	8956	33184	42140
Rural Stores	7	658	665

Notes: This table reports the number of unique UPCs, states, counties, and stores in both the treated and control states covered by the feminine hygiene product dataset used in this study. In addition to the full sample, the number of UPCs is separately reported for the subset of products only consumed by low-income households and for those only consumed by high-income households. The number of counties is further disaggregated into urban and rural counties. The number of stores is also broken down by store channel (convenience stores, drug stores, food stores, and mass merchandisers) and by location (urban and rural).

Source: NielsenIQ Retailer Scanner Data and Consumer Panel Data

Table B.2: Summary Statistics of Feminine Hygiene Products Quantity and Price

	Before Tax Policy			After Tax Policy		
	(1)	(2)	(3)	(4)	(5)	(6)
	Treated Group	Control Group	SMD	Treated Group	Control Group	SMD
Panel A: Quantity						
All Channels	22302.270	18431.070	0.118	18806.910	15272.100	0.117
Convenience Stores	114.098	131.262	-0.197	228.662	246.669	-0.012
Drug Stores	17838.310	11402.530	0.589	16728.940	10602.610	0.583
Food Stores	26533.940	22880.560	0.104	27564.910	22653.470	0.145
Mass Merchandiser	29076.990	23441.840	0.120	29663.760	23305.230	0.138
Urban Stores	22324.700	18717.560	0.110	18823.190	15494.270	0.110
Rural Stores	3863.671	4624.243	-0.260	4287.438	4010.820	0.094
Panel B: Price						
All Channels	4.883	4.607	0.218	5.171	4.854	0.230
Convenience Stores	4.596	4.570	0.079	4.427	4.391	0.057
Drug Stores	5.595	5.538	0.106	6.163	6.179	-0.026
Food Stores	5.097	4.646	0.303	5.570	4.973	0.388
Mass Merchandiser	3.652	3.687	-0.034	3.874	3.811	0.055
Urban Stores	4.884	4.618	0.210	5.172	4.868	0.221
Rural Stores	3.685	4.041	-0.380	3.874	4.158	-0.278

Notes: This table presents the average monthly total sales quantity per store in treated and control states, both before and after the implementation of the tax policy. In addition to the full sample, the summary statistics are provided separately by store channel—convenience stores, drug stores, food stores, and mass merchandisers—as well as by location type (urban versus rural stores). Panel A reports results for sales quantities, while Panel B reports results for sales prices.

Source: NielsenIQ Retailer Scanner Data

Table B.3: Stacked Difference-in-difference Results of Heterogeneous Analysis

	(1)	(1)	(1)	(1)	(1)	(1)
	Convenience	Drug	Food	Mass	Urban	Rural
Panel A: Quantity						
Treat ✕ Post	0.023**	0.020**	0.000	0.018***	0.021**	-0.004
	(0.009)	(0.010)	(0.009)	(0.005)	(0.010)	(0.004)
State×Cohort FE	✕	✕	✕	✕	✕	✕
Time×Cohort FE	✕	✕	✕	✕	✕	✕
UPC×Store FE	✕	✕	✕	✕	✕	✕
Number of Observations	2,033,582	138,017,889	131,321,735	93,905,602	361,485,567	3,790,973
Panel B: Price						
Treat ✕ Post	0.003	-0.003	0.012	0.000	0.001	0.014
	(0.009)	(0.005)	(0.003)	(0.002)	(0.005)	(0.015)
State×Cohort FE	✕	✕	✕	✕	✕	✕
Time×Cohort FE	✕	✕	✕	✕	✕	✕
UPC×Store FE	✕	✕	✕	✕	✕	✕
Number of Observations	2,033,582	138,017,889	131,321,735	93,905,602	361,485,567	3,790,973

Notes: This table reports the results from a stacked difference-in-differences analysis examining heterogeneous effects. Columns (1) through (4) present estimates based on samples restricted to convenience stores, drug stores, food stores, and mass merchandisers, respectively. Panel A reports the regression results for sales quantity, while Panel B reports the results for sales prices.

Source: NielsenIQ Retailer Scanner Data

Table B.4: Public Finance Implications of Sales Tax Exemption vs. Revenue Equivalent Means-Tested Cash Grant

State	Menstrual Age Females	Menstrual Age Females Below FPL	Sales Tax Rate	Revenue Lost from Repeal	Revenue Loss per Female	Equivalent Cash Grant to FPL Female
Alabama	1,270,887	218,593	4.80%	$4,534,530	$3.57	$20.74
Arizona	1,806,351	242,051	5.60%	$7,519,249	$4.16	$31.06
Arkansas	754,273	129,735	6.50%	$3,644,401	$4.83	$28.09
California	9,985,338	1,298,094	7.25%	$53,812,723	$5.39	$41.46
Colorado	1,521,246	153,646	2.90%	$3,279,304	$2.16	$21.34
Connecticut	869,942	98,303	6.35%	$4,106,277	$4.72	$41.77
Florida	5,242,070	691,953	6.00%	$23,379,661	$4.46	$33.79
Georgia	2,889,527	424,760	4.00%	$8,591,537	$2.97	$20.23
Hawaii	330,152	35,326	4.00%	$981,654	$2.97	$27.79
Idaho	490,851	54,975	6.00%	$2,189,197	$4.46	$39.82
Illinois	3,128,030	397,260	6.25%	$14,532,325	$4.65	$36.58
Indiana	1,690,288	233,260	7.00%	$8,795,141	$5.20	$37.71
Iowa	770,931	96,366	6.00%	$3,438,356	$4.46	$35.68
Kansas	721,956	88,801	6.50%	$3,488,257	$4.83	$39.28
Kentucky	1,093,823	200,170	6.00%	$4,878,457	$4.46	$24.37
Louisiana	1,140,993	240,749	4.45%	$3,774,218	$3.31	$15.68
Maine	314,888	36,212	5.50%	$1,287,370	$4.09	$35.55
Maryland	1,540,346	161,736	6.00%	$6,869,952	$4.46	$42.48
Massachusetts	1,759,975	202,397	6.25%	$8,176,561	$4.65	$40.40
Michigan	2,396,695	349,917	6.00%	$10,689,274	$4.46	$30.55
Minnesota	1,399,549	142,754	6.88%	$7,152,289	$5.11	$50.10
Mississippi	744,251	145,873	7.00%	$3,872,590	$5.20	$26.55
Missouri	1,514,567	196,894	4.23%	$4,756,629	$3.14	$24.16
Nebraska	488,428	54,216	5.50%	$1,996,859	$4.09	$36.83
Nevada	788,684	100,952	6.85%	$4,015,854	$5.09	$39.78
New Jersey	2,255,336	243,576	6.63%	$11,106,602	$4.92	$45.60
New Mexico	512,269	98,356	5.00%	$1,903,936	$3.72	$19.36
New York	4,846,842	746,414	4.00%	$14,411,297	$2.97	$19.31
North Carolina	2,710,337	379,447	4.75%	$9,569,760	$3.53	$25.22
North Dakota	192,766	22,168	5.00%	$716,449	$3.72	$32.32
Ohio	2,836,740	417,001	5.75%	$12,124,714	$4.27	$29.08
Oklahoma	1,018,737	177,260	4.50%	$3,407,679	$3.35	$19.22
Pennsylvania	3,078,959	406,423	6.00%	$13,732,176	$4.46	$33.79
Rhode Island	269,599	34,239	7.00%	$1,402,816	$5.20	$40.97
South Carolina	1,311,940	200,727	6.00%	$5,851,260	$4.46	$29.15
South Dakota	216,461	27,707	4.50%	$724,063	$3.35	$26.13
Tennessee	1,778,100	275,606	7.00%	$9,252,061	$5.20	$33.57
Texas	8,059,130	1,208,869	6.25%	$37,441,422	$4.65	$30.97
Utah	941,368	89,430	6.10%	$4,268,483	$4.53	$47.73
Vermont	148,299	15,127	6.00%	$661,415	$4.46	$43.73
Virginia	2,184,956	249,085	5.30%	$8,608,008	$3.94	$34.56
Washington	1,964,308	227,860	6.50%	$9,490,894	$4.83	$41.65
West Virginia	402,883	74,130	6.00%	$1,796,860	$4.46	$24.24
Wisconsin	1,402,009	162,633	5.00%	$5,210,806	$3.72	$32.04
Wyoming	138,486	17,588	4.00%	$411,767	$2.97	$23.41

Notes: "Menstrual Age Female" is estimated females between age 12.5 and 51 years. Assumes menstrual age females spend $74.33 annually on feminine hygiene products eligible for state sales tax exemption. Total tax expenditure is the amount of state sales tax revenue forgone by the exemption. Sales tax savings per female represents the avoided sales tax payments per all menstrual age females. Targeted grant per female is the tax expenditure equivalent program for a cash grant targeting menstrual age females in poverty. These results presented on Figure 4 and discussed in section 5. See section A for sources and calculation details.